PAPER_898: Phonon Lagrangian $\Phi \cdot S_{26}$ Derivation

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** PhononLagrangianPhiS26DerivationCalc (CP4 #482) **CVW:** v2.0.0 compliant

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Abstract

Complete phonon-modulated Lagrangian L_phonon = $E\_net \cdot V \cdot \Phi \cdot S_{26}$ with Euler-Lagrange variation $\delta S/\delta\varphi$_phonon = 0. Incorporates Kozima coupling in the phonon regime and provides the variational closure for phonon-enhanced UQFF dynamics.

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1. Core Equations

```
L_phonon = E_net·V·Φ_{1.25THz}·S₂₆
δS/δφ_phonon = 0
```

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2. Parameters

Parameter	Default	Description
E_0	1.0 J	Initial energy scale
t	0.0 s	Time parameter
V_filament	1e48 m^3	Filament volume
F_UBi_over_FU	0.8	Buoyancy-to-field ratio
omega	$2\pi\times1.25e12$ rad/s	Angular frequency

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3. UQFF Integration

This calculator integrates et_phonon_resonance.py from Session 208 into the CP4 pipeline as class #482.

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